# Basic (trace the loop on paper) — Hints

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Only 2 hints — work through them in order, and don't jump ahead until you've genuinely tried. Each hint builds toward the full solution.


- Hint 1 — The idea, and picking the right task
- Hint 2 — Writing the full trace


## Hint 1 — The idea, and picking the right task

The task has to be genuinely sequential, not just "2 things to do." "What's the weather in Paris, and what's the weather in Tokyo?" is 2 independent calls — either could go first, or even run at once. "What's the weather in Paris, in Fahrenheit?" is different: you cannot call the conversion tool until you already have the Paris temperature, because its input *is* the first tool's output. That dependency is what makes this a *loop* instead of just "2 tool calls" — write your task, then check it has this property before tracing anything.

Two traps worth planning for once you're comfortable with the basic shape: **a task that looks like it needs a tool but doesn't** (e.g., "what's 15% of the Paris temperature, if it's 20°C?" — once you already know it's 20, the arithmetic needs no tool at all), and **a tool that could plausibly fail** (what would the model's next Thought look like if `get_weather("Paris")` returned an error instead of a temperature?). Just notice, before picking your task, that "which tool, and when" is a real decision the model has to get right, not a given — these are the 2 failure modes the rest of this document is built around.

## Hint 2 — Writing the full trace

Be precise about *why* each Thought leads to that specific Action — this is the part people skip, and it's the part that actually builds the mental model. "Thought: I need the current temperature in Paris before I can convert it, so I can't answer yet" names what's missing and why the next Action follows from it. "Thought: I'll check the weather" with no reasoning attached does not. Now trace a second task — "Convert 100 US dollars to euros, then tell me if that's enough for a 90-euro hotel room" — twice: once the correct way (1 tool call, then reasoning over numbers you already have), and once the wasteful way (an unnecessary second "compare numbers" tool call). Comparing your 2 traces side by side is the point, not just producing 1 correct one.

One more worth tracing before you check the Solution: the weather tool call from round 1 fails (`get_weather("Paris")` returns `"Error: service unavailable"` instead of a temperature). Write the Thought that should follow a failed Observation — does the model retry, try something else, or give up and say so honestly? This is deliberately hard to get "right" on paper; the point is noticing that a failed Observation needs its own Thought, not something the loop can just skip over as if it never happened.

Full solution: Show me the solution